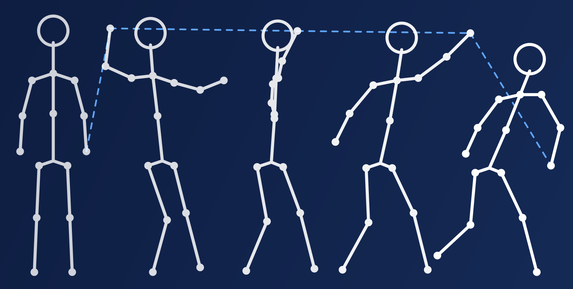
ESA-GCN

Efficient Semantic-Aware Graph Convolution for Skeleton-Based Action Recognition

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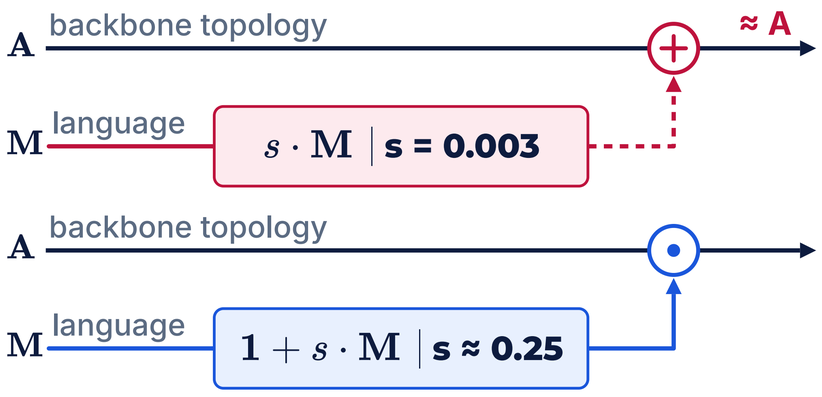
Multiply, don't add. Learn ε, don't search it.

25 / 25	+0.53 to +0.73 pp	4 of 5 splits	≤ +2.6% params
beat retrained LA-GCN	4-stream gain on every split	best or tied among 13 compared	added cost, with +4.9% FLOPs

Additive gates are prone to collapse

Additive $\mathbf{A} + s \cdot \mathbf{M}$

Ours $\mathbf{A} \odot (1 + s \cdot \mathbf{M})$



An additive gate's s drifts to 0.003. SEM starts as identity, and s grows from 0.1 to 0.22–0.29 on every split.

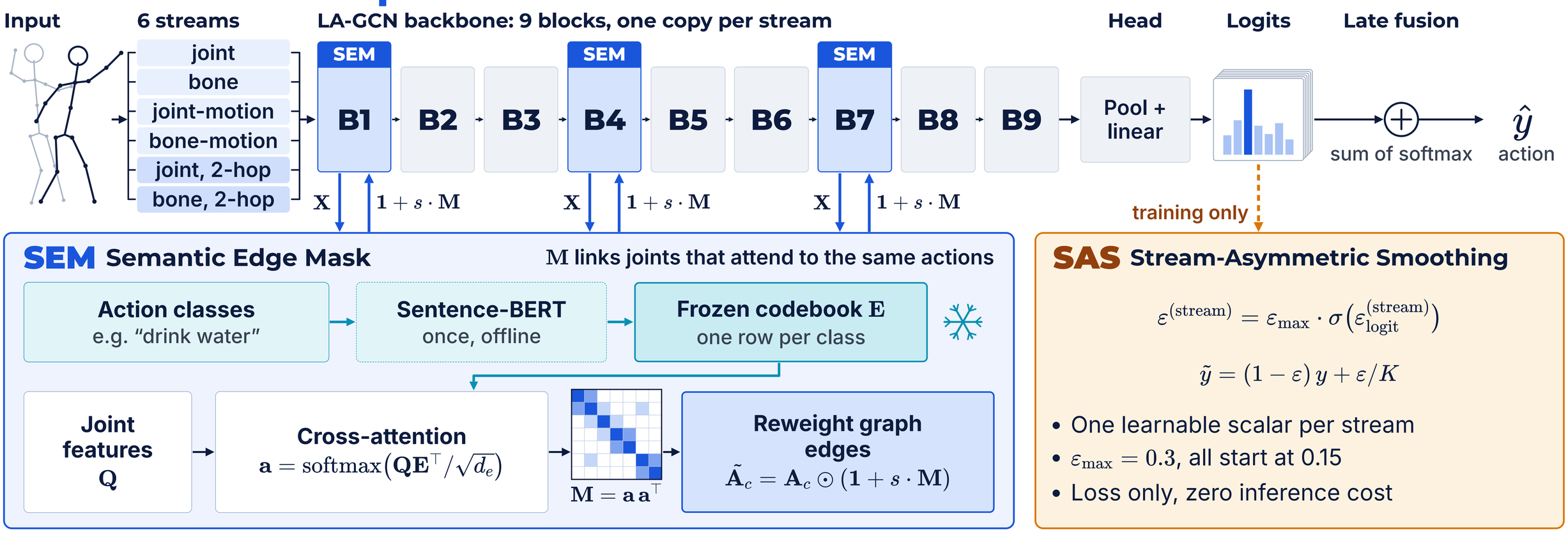
Label smoothing is grid-searched

0.05 joint, joint-motion

0.20 bone, bone-motion

Each modality needs its own smoothing ε, found by grid search. **A fourfold gap that each stream can learn instead.**

ESA-GCN: two compact modules on the LA-GCN backbone



Best or tied on 4 of 5 splits

Six-stream top-1 (%) among the 13 methods compared in the paper.

Method	Params (M)	NTU 60 X-Sub	NTU 60 X-View	NTU 120 X-Sub	NTU 120 X-Set	NW-UCLA
InfoGCN	1.57	93.0	97.1	89.8	91.2	97.0
HD-GCN	1.66	93.4	<u>97.2</u>	90.1	91.6	97.2
LA-GCN	1.44	<u>93.5</u>	<u>97.2</u>	**90.7**	<u>91.8</u>	<u>97.6</u>
ESA-GCN	1.47	**93.65**	**98.02**	<u>90.19</u>	**92.17**	**97.63†**

Comparators quoted from their papers. **Bold** best, <u>underlined</u> second. † Ties 97.6. Higher than InfoGCN and HD-GCN on every split, with fewer parameters per stream.

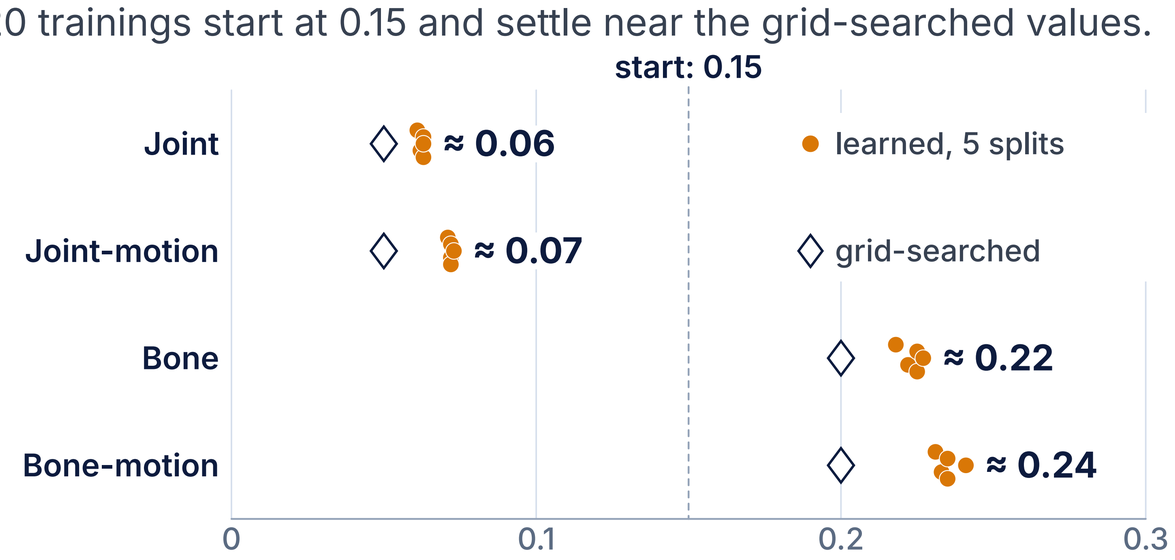
25 / 25 beat retrained LA-GCN

	Joint	Bone	J-motion	B-motion	4-stream
60 X-Sub	+0.35	+0.81	+0.50	+1.34	+0.53
60 X-View	+0.53	+0.81	+0.50	+1.16	+0.53
120 X-Sub	+0.64	+0.81	+0.66	+1.37	+0.73
120 X-Set	+0.63	+0.77	+0.70	+1.37	+0.73
NW-UCLA	+0.64	+1.94	+0.44	+2.15	+0.64

Top-1 gain (pp) over LA-GCN retrained in our pipeline and schedule. Largest on bone streams, up to +2.15.

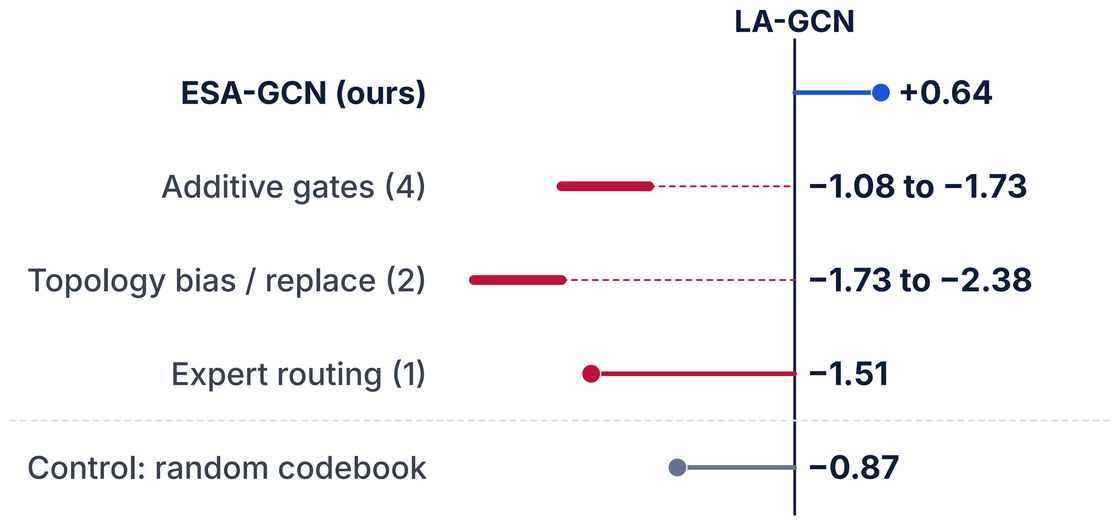
Learned ε recovers the 4× gap

20 trainings start at 0.15 and settle near the grid-searched values.



Multiply beats 7 alternatives

NW-UCLA joint, change in top-1 (pp) vs LA-GCN.



Take-home

Inject language by multiplying the skeleton graph, not adding to it, and let each stream learn its own label smoothing. Two compact modules beat a retrained LA-GCN on all 25 results for ≤ 2.6% more parameters.

Xu et al., LA-GCN, IEEE TMM 2025. Chen et al., CTR-GCN, ICCV 2021. Reimers and Gurevych, Sentence-BERT, EMNLP 2019.

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